FX-JEPA

Learning Effect-Aware Audio Representations for Reference-Based Audio Effect Transfer

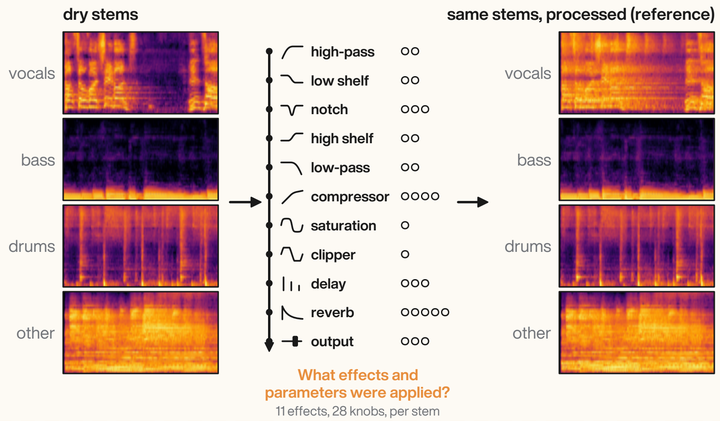
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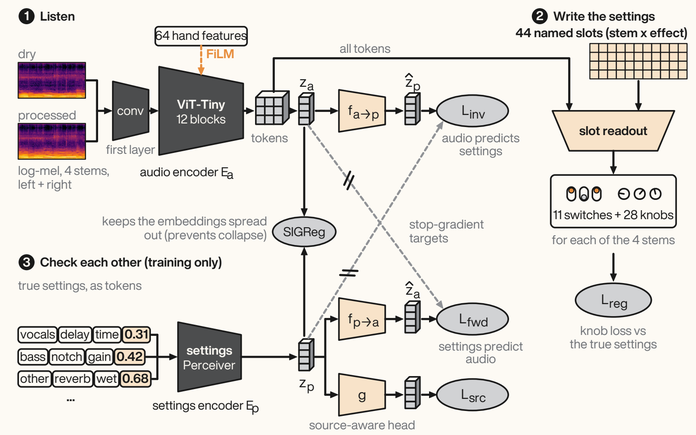
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Reference-based effect style transfer

FX-JEPA predicts the strip settings in latent space, then renders once.

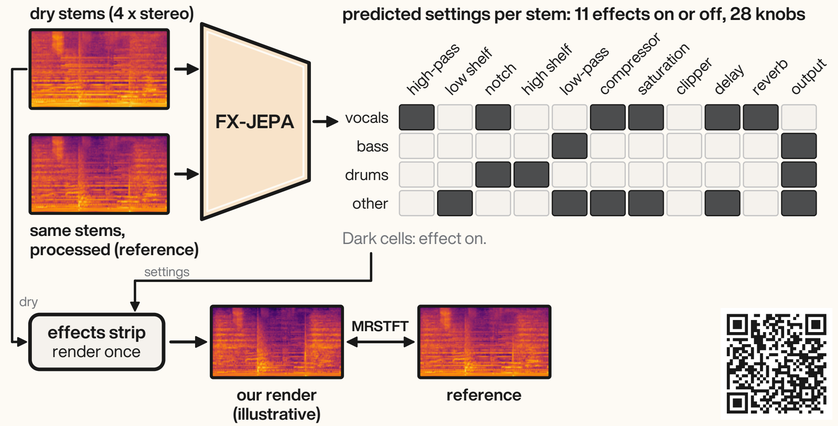


The FX-JEPA model



$$\mathcal{L} = \frac{1-2\lambda}{3}(\mathcal{L}_{inv} + \mathcal{L}_{fwd} + \mathcal{L}_{src}) + \lambda(S_a + S_p) + w_D D + \mathcal{L}_{reg}$$

$\lambda = 0.02$, $w_D = 0.1$. $\mathcal{L}_{inv}$: audio predicts settings. $\mathcal{L}_{fwd}$: settings predict audio. $\mathcal{L}_{src}$: settings plus the dry stems predict the pair of each song. $\mathcal{L}_{reg}$: the knob loss. S_a, S_p: SIGReg, keeps each branch spread. D: the pairs of one song stay near their mean.



Audio examples
dry, reference, ours; 8 clips

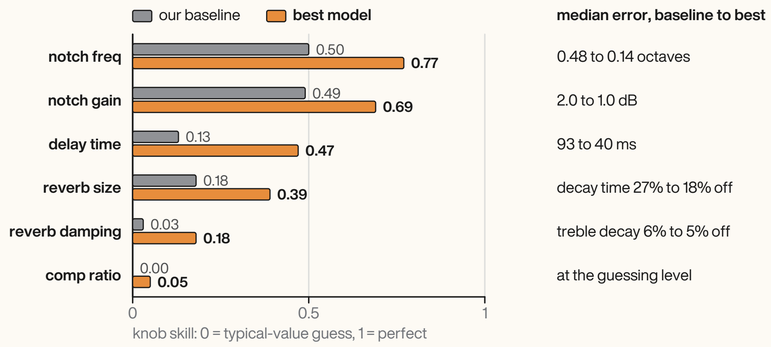
65%
less spectral distance than leaving the stems dry

0.1 s
to predict one clip's settings, on one GPU

10.9%
less spectral distance than our baseline, from the slot readout alone

Ablations

The baseline predicts all 44 settings from one averaged vector. FX-JEPA gives each stem and effect its own learned query over the audio tokens. That change alone cuts the spectral distance by 10.9%, encoder unchanged.



median error, baseline to best
0.48 to 0.14 octaves
2.0 to 1.0 dB
93 to 40 ms
decay time 27% to 18% off
treble decay 6% to 5% off
at the guessing level

Spectral distance to the reference

MRSTFT, 1,000 held-out clips from 929 songs; lower is better.

No processing
the dry stems, unchanged — 1.17

Best published encoder, as released
best of four: StemFX, AFx-Rep, FX-Encoder++, RelFX; frozen, small readout on top — 0.869

StemFX design, retrained on our data
our data and recipe — 0.485

FX-Encoder++ design, retrained on our data
our data and recipe — 0.463

Our baseline
ViT-Tiny; all settings from one averaged vector; second run 0.504 — 0.492

Baseline + conv first layer
convolutions before the ViT; alone, within run-to-run noise — 0.495

Baseline + slot readout
one query per stem and effect — 0.439

Both changes (FX-JEPA)
conv first layer + slot readout — **0.413**

0 — 0.5 — 1.17
MRSTFT, 0 = identical to the reference; lower is better

Terms

slot readout
one learned query per stem and effect, 44 in all; each reads the audio tokens and predicts that effect's settings

conv first layer
a small convolution stack in place of the ViT's patch embedding, at the input

JEPA
joint embedding predictive architecture; audio and settings embeddings predict each other

knob loss
regression loss against the true switches and knob values

data FMA and MoisesDB stems; public data only; 44.1 kHz, 10 s clips

effects re-implemented from open-source parts: five-band EQ as in MultiAFx, compressor from torchcomp (Yu et al., 2024), tanh saturation, hard clipper, feedback delay, Freeverb reverb; fixed order